

MINJI SON

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RESEARCH INTERESTS

Resource management and performance predictability for ML workloads on shared GPU infrastructure — GPU scheduling in multi-tenant containerized clusters; systems support for ML inference serving with predictable tail latency under co-location.

EDUCATION

Sungkyunkwan University, South Korea

Mar 2015 – Aug 2019

B.S. in Software Engineering, Department of Computer Science and Engineering

- GPA: 3.72 / 4.5 (92.2 / 100); Major GPA: 3.93 / 4.5
- Academic Excellence Scholarship; Dean's List Award
- Relevant Coursework: Operating Systems, Compiler Design, Computer Architectures, Computer Networks, System Programming, Introduction to Database, Introduction to Artificial Intelligence

RESEARCH EXPERIENCE

Undergraduate Research Assistant, ARCS Lab, Sungkyunkwan University

Mar 2017 – Dec 2018

Advisor: Prof. HwanSoo Han (deep-learning compilers, big-data analytics systems, and system software)

GPU Performance Characterization on Containers and Multi-Task Scheduling (First author, KSC 2018)

- Designed and led a three-part experimental study of NVIDIA Docker across 19 Rodinia and Polybench/GPU benchmarks on Maxwell (GTX 980 Ti) and Volta (V100) GPUs
- Isolated the container software-stack overhead as a constant ~ 0.96 s independent of workload, establishing that GPU containerization cost does not scale with application runtime
- Devised a cache-sensitivity classification — kernels whose L1 hit rate rises $>10\%$ under a doubled effective cache capacity — to predict when MPS-based GPU sharing helps vs. hurts, showing MPS accelerates co-located execution until cache saturation, after which performance inverts
- Demonstrated hard device-memory limits under container co-location and proposed unified memory (Pascal+) as a mitigation

Performance Analysis of Spark Applications through Task Execution (KSC 2017)

- Co-developed the Task Impact Factor (TIF) metric quantifying each RDD operation's share of end-to-end job time across executor configurations and 25–100 GB inputs
- Analyzed a performance-inversion effect where higher core counts degrade throughput once input exceeds memory, tracing the cause to JVM garbage-collection pressure

PUBLICATIONS

- **MinJi Son**, HyunJun Kim, and HwanSoo Han, "Analyzing GPU Performance on Containers and Characteristics of Multi-Task Scheduling," Korea Software Congress (KSC), 2018.
- MinSeop Jung, DongEun Lee, **MinJi Son**, JeongHwan Park, and HwanSoo Han, "Performance Analysis of Spark Application through Task Execution," Korea Software Congress (KSC), 2017.

WORK EXPERIENCE

Samsung Electronics, Mobile eXperience Division — Software Engineer

Jul 2021 – Present

One of three HQ engineers developing core components of the Samsung Messaging Platform (SMP) — a globally distributed push-notification relay infrastructure serving 1.26B Galaxy devices with 600M+ daily pushes via FCM, while leading technical design for a seven-engineer team in Vietnam.

Distributed data-warehouse performance engineering (Amazon Redshift, MPP)

- Diagnosed a **5–10x** performance regression in a multi-billion-row data warehouse serving audience-targeting workloads, and reduced worst-case query latency from 6.5 min (timeout) to under 1 min (**7–13x**)

- Traced the regression to interacting root causes across system layers: lookup-join cardinality amplification, a distribution-key/join-key mismatch, and a 31x cardinality misestimation from stale table statistics — compounded by sort-order decay degrading zone-map pruning on a 3.3B-row table
- Redesigned distribution/sort keys and join ordering to achieve colocated (zero-redistribution) joins, verifying that every optimization preserved result counts via execution-plan diffs
- Accelerated a nightly per-service statistics batch by precomputing an expensive shared join as materialized views — applied selectively to large services after measuring that MV overhead regressed small ones — roughly halving total batch time ($\sim 2.6 \text{ h} \rightarrow \sim 1.2 \text{ h}$) with verified output parity

ML inference integration and service reliability

- Integrated an AI moderation service (multimodal LLM, OCR, object detection) into SMP's content-approval pipeline, designing asynchronous request handling and fallback-to-human routing — automating 87% of content-submission reviews in production evaluation
- Configured HPA for core SMP services on Kubernetes (EKS) to absorb device- and FCM-traffic surges, hardening scale-in with graceful termination so pod removal drains in-flight push requests without loss

Korea Telecom, Connected Car Platform Development Team — Software Engineer Jul 2019 – Jul 2021

- One of two engineers owning the third-party integration layer of a global automotive OEM's membership service in production — designed, built, and operated backend integrations end-to-end, including API contract coordination with external providers and production incident response

Samsung Electronics, Device Solutions — Intern

Jul 2018 – Aug 2018

- Enhanced SSD failure-analysis tooling with automated diagnostic summarization and a streamlined dump-loading process

SELECTED PROJECTS

Cloud and Server Starter — Samsung Electronics

Aug 2024 – Mar 2025

- Selected internal engineering challenge (team of 5); built a Terraform IaC service provisioning AWS/Azure infrastructure — internal security policies enforced by default — in under 1 hour (previously days); owned compute and load-balancing modules

PATENTS

Co-inventor on 7 patent applications (KR/US/EP, Samsung Electronics, 2023) in on-device ML applications including handwriting recognition and content filtering.

TEACHING EXPERIENCE

Teaching Assistant, Compiler Design (Prof. HwanSoo Han), Sungkyunkwan University Spring 2018

TA, Computational Thinking and SW Coding, Sungkyunkwan University Spring 2017

PROFESSIONAL ACTIVITIES

WITS (Women in Technology at Samsung), Samsung Electronics

Jan 2025 – Dec 2025

- Selected for a company-wide mentoring program with senior women engineering leaders

WISET Engineering Research Program, Korea Center for Women in Science

Apr 2017 – Oct 2017

- Selected for a competitive, government-funded research program (100 teams funded nationwide)
- Investigated the performance of zswap (Linux compressed swap) in mobile environments and techniques for improving response time, as part of a 5-member research team

SKILLS

- **Systems & Infrastructure:** Linux, Docker, Kubernetes, Terraform, CUDA, Apache Spark
- **Cloud (AWS):** EKS, Redshift, EC2, Lambda, CloudWatch
- **Programming & Backend:** Java, C, C++, Python; Spring Boot, Spring Batch, MySQL, PostgreSQL
- **Languages:** Korean (native), English (fluent, TOEFL 110), Japanese (conversational)